Shrey Painuli

Roll No.: M25EET007
B.Tech - Electrical Engineering
M.Tech - Sensors and Internet of Things
Indian Institute of Technology Jodhpur

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Github
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech	Indian Institute of Technology Jodhpur	9.25	2025-Present
B.Tech	G.B Pant Institute of Engineering and Technology, Pauri	84.26	2017-2021
Senior Secondary	ICSE Board	82.25	2017
Secondary	ICSE Board	94.40	2015

Relevant Coursework (M.Tech)

- Digital VLSI Design
- Hardware Design for AI
- Hardware Software Co-design
- Embedded Systems Design
- Micro-systems Fabrication
- Data Communication & Networking

Experience

- **Chegg India Pvt. Ltd.** *Jan. 2021 - Dec. 2022*
  Subject Matter Expert Work From Home

Projects

- **R4MDC FFT Hardware Accelerator** *Github*
  RTL Design, Simulation, FPGA Synthesis and Implementation
  – Designed a high-throughput pipelined FFT accelerator using Radix-4 Multi-Delay-Commutator architecture in Verilog, with streaming datapath, butterfly units, and complex twiddle multipliers for continuous processing.
  – Optimized microarchitecture using deep pipelining and parallelism; verified functionality and achieved 17-cycle first-output latency and 36.36M FFT frames/s throughput at 100 MHz.
- **Hardware-efficient kNN AI Accelerator** *Github*
  RTL Design, Functional Simulation, FPGA Synthesis and Implementation
  – Designed a hardware-efficient kNN inference engine in Verilog with Euclidean, Manhattan, and Chebyshev distance metrics using a common pipelined datapath for real-time AI classification.
  – Performed functional and post-implementation simulation; analyzed FPGA area, power, and thermal trade-offs, identifying low-power alternatives for hardware-aware AI.
- **Design and Verification of APB RAM** *Github*
  RTL Design and SystemVerilog Verification
  – Designed an APB-compliant memory-mapped RAM in Verilog with address decoding, read/write control, and handshake-based data transfer for SoC integration.
  – Developed a modular SystemVerilog testbench with generator, driver, monitor, and scoreboard to verify functional correctness using constrained-random transactions.
- **Navigation and Object Identification Aid for the Visually Challenged** *Github*
  B.Tech Major Project, Mentor: Prof. Sanjay Gairola, H.O.D-GBPIET
- **Exigence Band** *Github*
  B.Tech Minor Project
- **Ethernet Integration and Cache Coherence Implementation in Dual-Core Cheshire RISC-V SoC**
  Ongoing Project, Mentor: Dr. Binor Kumar, IIT Jodhpur

Technical Skills

- **Programming Languages**: Verilog, SystemVerilog, C/C++, Embedded C, Python, Tcl, BaSh, MATLAB, Java
- **Developer Tools**: Xilinx Vivado, Icarus Verilog, Git, Linux, Visual Studio, Keil MDK

Extracurriculars

- II place, Fresher's Volleyball Tournament (IIT Jodhpur)
- Literary and Editorial Head (S.E.E, G.B.P.I.E.T)
- I Place, Josh Basketball Competition (G.B.P.I.E.T)
- III place, Technical Quiz (S.E.E, G.B.P.I.E.T)
- International Robo Soccer (Technoxian)
- III place, Mr. Goonj (College Fest Title, G.B.P.I.E.T)

Publications

- **Painuli, S.,** Gairola, S., Arya, K., Tiwari, D., Balram. (2021). Navigation and Object Identification Aid for the Visually Challenged. SAMRIDDHI : A Journal of Physical Sciences, Engineering and Technology, Volume 13, Special Issue (2), 168-173 *Research Publication*